LEARNING MODULE

13

Probability Foundations and Bayes

A frequency becomes a probability: sample spaces, conditioning, independence and Bayes' rule, worked on two trades of a bet and on 5,849 TAIEX days.

VOLUME III · Probability **BUILDS ON** LM1 · LM3 · LM4 · LM5

LEADS TO LM14 · LM15 · LM16 · LM17 · LM18 · LM21 · LM24 · LM26 **TIME** ≈ 2.5 h + lab

LEARNING OUTCOMES

Mastery	*After this module you should be able to:*
☐	1 describe a random experiment by its sample space and events, and combine events with complements, unions and intersections
☐	2 derive the complement rule, monotonicity and inclusion–exclusion from the axioms of probability, and apply them to equally likely outcomes and to the empirical distribution of a return sample
☐	3 calculate conditional probabilities from counts and apply the multiplication rule to consecutive trading days
☐	4 decide whether events, trials or discrete random variables are independent, distinguish pairwise from mutual independence and statistical from linear independence, and compute the probability of at least one event in independent trials
☐	5 apply the law of total probability and Bayes' rule with base rates, counts and odds, and explain why a conditional probability and its inverse can differ several times over
☐	6 define a random variable as a function on the sample space, and find the probability mass function and distribution function of a discrete random variable, an indicator included
☐	7 compute conditional frequencies, Bayes' rule and independence checks from a return series in Python, counting threshold days with a tolerance for exact ties

LEARNING MODULE OVERVIEW

- A probability model has a sample space Ω, events (subsets of Ω) and a probability that obeys three axioms. The complement rule, monotonicity, inclusion–exclusion (13.1) and the union bound (13.2) follow from them.
- The empirical distribution of a sample, mass $1/T$ on each day, is a probability model. Every frequency of TR23, the 5,849 daily returns of the TAIEX total-return index from 3 January 2003 to 8 October 2026, is an exact probability of a day drawn at random from it, not a forecast.
- Conditioning shrinks the sample space, (13.4). Of the 489 days after a move of 2% or more, 32 fell 3% or more: 6.54%, against 1.85% of all days. The same 32 days are 29.63% of the 108 falls.
- Independence is a product rule, (13.6). A down day followed a down day 1,226 times in 5,848 day pairs, against 1,195.0 if days were independent; two falls of 3% or more in a row came 13 times, against 2.0.
- For independent days, (13.7) puts at least one fall of 3% or more in 31.1% of blocks of 20 sessions. TR23's falls cluster, and only 20.9% of its 292 blocks hold one.
- Total probability (13.8) recombines the 7% and 10% limit regimes, with falls on 2.34% and 1.30% of their days, into 1.85%. Bayes' rule (13.9) turns a base rate of 1.847%, a hit rate of 29.63% and a false-alarm rate of 7.962% into 6.54%; in odds, (13.10), $0.018815 \times 3.7215 = 0.07002$.
- Pairwise independence is not mutual independence, disjoint events with positive probabilities are never independent, and statistical independence is not the linear independence of LM9.
- A random variable is a function on Ω. Two trades of LM4's +2%/−1% bet at 25 times equity end at 2.25, 1.125 or 0.5625 times the start, with probabilities 1/4, 1/2 and 1/4. Its distribution function counts the point itself: $F_X(1.125) = 3/4$, while $\Pr(X < 1.125) = 1/4$.

1 INTRODUCTION

Volumes I and II counted days: LM5 compared the share of 0050's days at or below −5% with a normal model, and LM3 averaged returns over a sample. This module turns such a share into a probability. A probability model has three parts: outcomes, the events built from them, and a rule that gives each event a number.

MOTIVATING CASE **After a big move**

The TAIEX total-return index, dividends reinvested, gives 5,849 daily returns from 3 January 2003 to 8 October 2026. On 108 days it fell 3% or more: 1.85% of days.

Big moves come in groups. Of the 108 falls, 32 (29.6%) came the day after a move of 2% or more either way, against 7.96% of the other days. Yet of the 489 days after such a move, only 32 fell 3% or more: 6.54%. That is 3.5 times the base rate, but still under one day in fifteen (Exhibit 1).

So "crashes tend to follow big moves" and "a big move rarely leads to a crash" are both true of the same 32 days. Why do the two shares differ, and how does Bayes' rule turn one into the other?

Exhibit 1: Falls of 3% or more and the day before, TR23

Yesterday's move	Fall of 3% or more today	Other days	All days	Share that fell
2% or more either way	32	457	489	6.54%
Smaller	76	5,283	5,359	1.42%
All	108	5,740	5,848	1.85%
Share after a big move	29.63%	7.96%	8.36%	

Note: the pair space, the 5,848 return dates with a previous return in the sample. Thresholds are inclusive, $|R_{t-1}| \geq 2\%$ and $R_t \leq -3\%$, and no return equals ±2% or –3%. The last row is the share of each column that followed a big move. Source: FinMind `TaiwanStockTotalReturnIndex` (TAIEX), retrieved 18 October 2026; cash dividends reinvested, no costs. Computations: `data/lm13_numbers.py` and `LM13_lab.ipynb`.

Section 2 states the axioms and shows that a return sample is a probability model. Sections 3 and 4 treat conditioning and independence. Section 5 derives Bayes' rule and resolves the case. Section 6 defines random variables, and Section 7 does the counting in Python.

NOTE **Notation for this module**

TR23 is the TAIEX total-return index: $T = 5{,}849$ daily simple returns R_t, total returns as in LM1 (cash dividends reinvested), 3 January 2003 to 8 October 2026. Here T is a count and t the date index.

The **pair space** is the 5,848 return dates from 6 January 2003 with a previous return; the first return, +2.24% on 3 January 2003, has none. Triples use the 5,847 dates from 7 January 2003. A **fall** is a day with $R_t \le -3\%$ and a **big move** a day with $|R_t| \ge 2\%$, thresholds inclusive.

Symbols: Ω, ω a sample space and an outcome (trades as strings of $+$ and $-$); A, B events named in each example (no annualizing here), A^c a complement; $\cup$, $\cap$, $\emptyset$, $\subseteq$ union, intersection, empty event, inclusion; $\#A$ the number of outcomes in A ($|\cdot|$ stays the absolute value); $A_1, \dots, A_m$ a list of m events or a partition, index i; $\mathbf{1}_A$, $\mathbf{1}\{\cdot\}$ indicators; X, Y, $X_1, \dots, X_n$ random variables, x, y, $x_1, \dots, x_n$ values; p_X, F_X pmf and distribution function, $\hat{F}$ TR23's empirical one; $x_{\text{lo}} < x_{\text{hi}}$ interval ends (a, b are the bet's); n independent trials, j wins, W_n wealth after n trades. The bet keeps LM4's p, $q = 1 - p$, b, a, f, f^*, g^*.

NOTE **Research hygiene**

Every frequency here is **descriptive** of TR23: the probability of a day drawn at random from the sample, not a forecast. Comparing the 7% and 10% limit regimes is a **stability check**. The independent-days figures are a **model** built from the same days. The bets and the manager are **hypothetical**. LM21 asks whether a gap is larger than chance.

2 SAMPLE SPACES, EVENTS AND THE AXIOMS

- [ ] describe a random experiment by its sample space and events, and combine events with complements, unions and intersections
- [ ] derive the complement rule, monotonicity and inclusion–exclusion from the axioms of probability, and apply them to equally likely outcomes and to the empirical distribution of a return sample

Start with an experiment small enough to list every outcome.

EXAMPLE 1

Two trades as a sample space

A trader takes LM4's +2%/−1% bet twice, at the Kelly position of 25 times equity. List the outcomes and the events "the first trade wins" and "at least one trade loses". When does wealth end below its start?

SOLUTION

Write $+$ for a win and $-$ for a loss. The **sample space** is $\Omega = \{++, +-, -+, --\}$. The first trade wins on $A = \{++, +-\}$, and some trade loses on $B = \{+-, -+, --\}$. Then $A^c = \{-+, --\}$, $A \cap B = \{+-\}$ and $A \cup B = \Omega$.

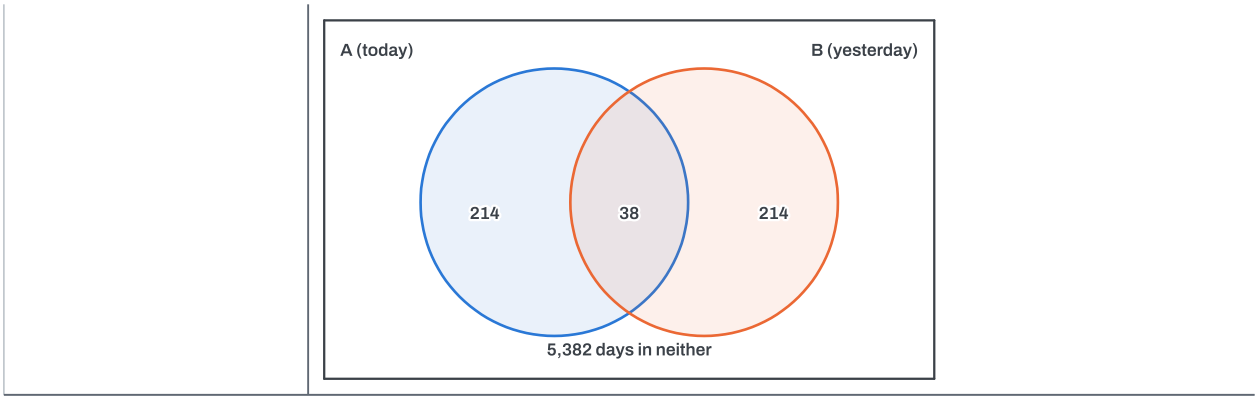
Suppose the four outcomes are equally likely, as for independent trades with $p = 1/2$ (Section 4). Then $\Pr(B) = 1 - \Pr(\{++\}) = 3/4$ and $\Pr(A \cap B) = 1/4$. At $f = 25$ a win multiplies wealth by $1 + 25 \times 2\% = 1.5$ and a loss by $1 - 25 \times 1\% = 0.75$. Since $1.5 \times 0.75 = 1.125$, wealth ends below its start only on $--$, at $0.75^2 = 0.5625$.

DEFINITION **Sample space, outcome, event**

The **sample space** Ω is the set of all outcomes ω of a random experiment. An **event** is a subset $A \subseteq \Omega$; it occurs when the outcome lies in A. The **complement** A^c holds the outcomes not in A, the **union** $A \cup B$ those in A or B, and the **intersection** $A \cap B$ those in both. Events with $A \cap B = \emptyset$, the empty event, are **disjoint**.

On the pair space, let A be "fell 2% or more today" and B "fell 2% or more yesterday". Exhibit 2 draws the two events.

Exhibit 2: Two events on the same days



Note: the 5,848 days of the pair space, with A the days down 2% or more and B the days after one, thresholds inclusive. The numbers count the days in each region; the areas are not to scale. By (13.1), $A \cup B$ holds $252 + 252 - 38 = 466$ days. Source: as Exhibit 1.

DEFINITION **Axioms of probability**

A **probability** on Ω assigns to each event A a number $\Pr(A)$ such that:

1. every event has $\Pr(A) \geq 0$;
2. the whole space has $\Pr(\Omega) = 1$;
3. disjoint events A and B have $\Pr(A \cup B) = \Pr(A) + \Pr(B)$.

Deep Dive 1 extends the third axiom to infinite sequences of disjoint events. Wasserman (2004, §1.3) takes that countable form as the third axiom.

Since A and A^c are disjoint with union Ω, the **complement rule** $\Pr(A^c) = 1 - \Pr(A)$ holds, and $\Pr(\emptyset) = 0$. If $A \subseteq B$, then B is the disjoint union of A and $B \cap A^c$, so $\Pr(A) \leq \Pr(B)$: **monotonicity**. Splitting $A \cup B$ into A and $B \cap A^c$, and B into $A \cap B$ and $B \cap A^c$, gives **inclusion–exclusion**:

$$\Pr(A \cup B) = \Pr(A) + \Pr(B) - \Pr(A \cap B). \tag{13.1}$$

Since $\Pr(A \cap B) \geq 0$, (13.1) gives $\Pr(A \cup B) \leq \Pr(A) + \Pr(B)$, and induction gives the **union bound**:

$$\Pr(A_1 \cup \cdots \cup A_m) \leq \sum_{i=1}^{m} \Pr(A_i). \tag{13.2}$$

When Ω is finite and its outcomes are equally likely, additivity gives each outcome the probability $1/\#\Omega$, so

$$\Pr(A) = \frac{\#A}{\#\Omega}. \tag{13.3}$$

A return sample is such a model: draw one of its T days at random, each with probability $1/T$. This is the **empirical distribution** of the sample. Under it, the probability of an event is the number of days in the event divided by T, so every frequency of the sample is an exact probability.

In this model every average is a sample average with divisor T, as in (3.7) of LM3; LM14 shows why. That a frequency settles near a probability as days accumulate is the law of large numbers of LM17.

EXAMPLE 2

Two falls of 2% or more

On the pair space, 252 days fell 2% or more (A), 252 followed such a fall (B), and 38 did both. Find $\Pr(A \cup B)$ and compare it with the union bound. On how many days would both occur if consecutive days were independent?

SOLUTION

By (13.1), $\Pr(A \cup B) = (252 + 252 - 38)/5{,}848 = 7.97\%$. Adding without the overlap counts 38 days twice and gives 504 days, 8.62%: the union bound (13.2), which is exact only when the overlap has probability 0.

Each event has probability $252/5{,}848 = 0.0431$. If consecutive days were independent (Section 4, a model), both would occur on about $5{,}848 \times 0.0431^2 = 10.9$ days, not 38.

PITFALL **Adding overlapping events**

"Or" adds probabilities only when the overlap has probability 0, as for disjoint events. The 8.62% of Example 2 counts 38 days twice. Use (13.1), or split the union into disjoint pieces first: $214 + 38 + 214 = 466$ days.

DEEP DIVE **Countable additivity and events of probability zero** optional · click to expand

Countable additivity extends the third axiom to a sequence of disjoint events $A_1, A_2, \ldots$:

$$\Pr\left(\bigcup_{i=1}^{\infty} A_i\right) = \sum_{i=1}^{\infty} \Pr(A_i).$$

Wasserman (2004, §1.3) takes this countable form as the third axiom.

Countable additivity gives **continuity**: if $A_1 \supseteq A_2 \supseteq \cdots$ decrease to A, then $\Pr(A_i) \to \Pr(A)$ (Wasserman 2004, Theorem 1.8). Indeed, each A_i is the disjoint union of A and the differences from $A_i \cap A_{i+1}^c$ on. So $\Pr(A_i)$ is $\Pr(A)$ plus the tail of one convergent series, which tends to 0. By complements, increasing events behave alike.

An event of probability zero need not be empty. Take infinitely many independent fair trades (Section 4), a model that exists (Durrett 2019, Theorem 2.1.21, stated without proof). "Every trade loses" lies inside "the first n trades lose", of probability 2^{-n} by (13.7). By monotonicity, its probability is at most 2^{-n} for every n, hence 0, although the outcome exists. "With probability one" names the complement of such an event; LM17 applies it to the wealth paths of LM4.

導讀 **樣本空間、事件與機率公設**

機率模型先列出所有可能的結果（outcome），全部結果構成樣本空間（sample space）Ω，事件（event）則是結果的集合，可以取餘事件（complement）、聯集（union）與交集（intersection）。機率公設（axioms of probability）只要求三件事：機率不為負、$\Pr(\Omega) = 1$、互斥（disjoint）事件的機率可以相加。餘事件規則（complement rule）、單調性（monotonicity）與 inclusion–exclusion，都由這三條推導出來。把加權股價報酬指數（TR23）的 5,849 個交易日各給 $1/T$ 的機率，就得到經驗分布（empirical distribution），樣本中的每一個頻率都是這個模型的精確機率。但這是「從樣本裡隨機抽一天」的機率，描述的是過去的樣本，不是對明天的預測。

KNOWLEDGE CHECK 1

If $\Pr(A) = \Pr(B) = 0.9$, how small can $\Pr(A \cap B)$ be?

ANSWER

By (13.1) and $\Pr(A \cup B) \le 1$, the intersection has probability at least $0.9 + 0.9 - 1 = 0.8$. Equality holds when $A \cup B = \Omega$: with ten equally likely outcomes, let A and B each miss a different one.

3 CONDITIONAL PROBABILITY

- [ ] calculate conditional probabilities from counts and apply the multiplication rule to consecutive trading days

TR23 was down on 2,644 of the 5,848 days of the pair space, 45.21%. Of the 2,643 days after a down day, 1,226 were down, 46.39%. The share changed because the set of days changed, and conditioning makes this precise.

DEFINITION **Conditional probability**

Let $\Pr(A) > 0$. The **conditional probability** of B given A is

$$\Pr(B \mid A) := \frac{\Pr(A \cap B)}{\Pr(A)}. \tag{13.4}$$

Conditioning on A shrinks the sample space to A and rescales by $\Pr(A)$, so that A gets probability 1. In the empirical model, (13.4) divides the days in $A \cap B$ by the days in A. For a fixed A, the conditional

probability obeys the three axioms: it is not negative, $\Pr(\Omega \mid A) = 1$, and disjoint events stay disjoint inside A.

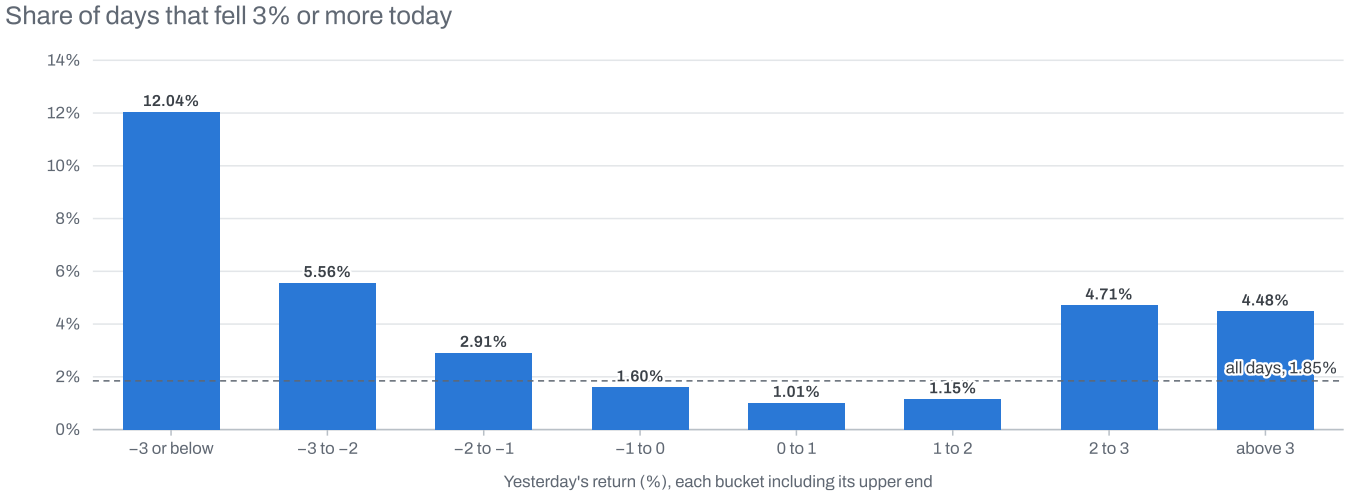
Multiplying (13.4) by $\Pr(A)$ gives the **multiplication rule**, and repeating it gives a chain:

$$\begin{aligned} \Pr(A \cap B) &= \Pr(A)\Pr(B \mid A), \\ \Pr(A_1 \cap \cdots \cap A_m) &= \Pr(A_1)\Pr(A_2 \mid A_1)\cdots \\ &\cdots \Pr(A_m \mid A_1 \cap \cdots \cap A_{m-1}). \end{aligned} \tag{13.5}$$

The chain needs $\Pr(A_1 \cap \cdots \cap A_{m-1}) > 0$. It telescopes, since each denominator cancels the numerator before it.

Exhibit 3 sorts the pair space by yesterday's return and shows, for each bucket, the conditional probability of a fall today.

Exhibit 3: A fall of 3% or more, by yesterday's return



Note: the pair space, 5,848 days. Days/falls per bucket, from the left: 108/13, 144/8, 516/15, 1,875/30, 2,272/23, 696/8, 170/8 and 67/3, which sum to 5,848 and 108. A bucket holds its upper end, as the intervals of LM5 do, and no return sits on an edge. Yesterday's returns end on 7 October 2026, so the bucket from −1% to 0 holds one day fewer than in Exhibit 7. Source: as Exhibit 1.

Both tails raise the probability of a fall: to 12.04% after a fall of 3% or more and to 4.48% after a rise of more than 3%, against 1.85% overall. Large moves come in clusters; LM26 models them.

EXAMPLE 3

Two falls in a row

On the pair space, 108 days follow a fall of 3% or more, and 13 of them fell 3% or more too. Find the probability of two falls in a row, and compare the conditional probability with the base rate.

SOLUTION

Let A_1 be "a fall yesterday" and A_2 "a fall today". Then $\Pr(A_1) = 108/5{,}848$ and $\Pr(A_2 \mid A_1) = 13/108 = 12.04\%$. By (13.5), $\Pr(A_1 \cap A_2) = 13/5{,}848 = 0.222\%$.

On the pair space a fall has probability $108/5{,}848 = 1.85\%$, so after a fall, a fall was 6.5 times as frequent. If days were independent (a model, Section 4), two falls in a row would occur on $108^2/5{,}848 = 2.0$ days, not 13.

EXAMPLE 4

Inside the tail

Of TR23's 5,849 returns, 108 are falls of 3% or more; 44 reached −4% and 15 reached −5%. Find $\Pr(R \le -4\% \mid R \le -3\%)$ and $\Pr(R \le -5\% \mid R \le -3\%)$, with R the return of a day drawn at random (Section 6).

SOLUTION

Since $\{R \le -5\%\} \subseteq \{R \le -3\%\}$, the intersection is the smaller event, and likewise for −4%. So (13.4) gives

$$\begin{aligned} &\Pr(R \le -5\% \mid R \le -3\%) \\ &\quad = 15/108 = 13.9\%, \\ &\Pr(R \le -4\% \mid R \le -3\%) \\ &\quad = 44/108 = 40.7\%. \end{aligned}$$

The 108 days averaged −4.06%, with divisor 108 as in LM3. Conditioning on $\{R \le -3\%\}$ gives a probability model on those days, a conditional distribution, and −4.06% is its average. Equation (5.11) of LM5 is the mean of such a conditional distribution; LM16 defines conditional expectation in general.

PITFALL **Small denominators**

A conditional probability is a ratio of counts, and its denominator can be small. The 12.04% of Exhibit 3 rests on 13 falls among 108 days and the 4.48% on 3 among 67. One more fall in each bucket would give 12.96% and 5.97%. Print the counts beside every conditional share.

導讀 **條件機率：把樣本空間縮小**

條件機率（conditional probability）$\Pr(B \mid A)$ 是把樣本空間（sample space）縮小到事件（event）A，再看其中有多少比例也屬於 B。在經驗分布（empirical distribution）中，它就是兩個計數的比：前一天漲跌 2% 以上的 489 天裡，有 32 天下跌 3% 以上，條件機率是 6.54%，而全部日子只有 1.85%。乘法規則（multiplication rule）把條件機率一個接一個相乘，可以算出連續兩天都大跌（跌 3% 以上）的機率。條件機率是比率，分母小的時候，多一天或少一天就會大幅改變結果，所以計數要和比率一起列出。

KNOWLEDGE CHECK 2

On the pair space, $\Pr(\text{down today} \mid \text{down yesterday}) = 1{,}226/2{,}643 = 0.4639$, and in reverse $\Pr(\text{down yesterday} \mid \text{down today}) = 1{,}226/2{,}644 = 0.4637$. Why do the two nearly agree, when the case's two conditional probabilities differ 4.5 times?

ANSWER

Both share the numerator 1,226, and their denominators, 2,643 and 2,644, almost match. By (13.4), $\Pr(B \mid A)/\Pr(A \mid B) = \Pr(B)/\Pr(A)$. In the case 489 days follow a big move but only 108 days fell, so $32/108$ is $489/108 = 4.53$ times $32/489$.

4 INDEPENDENCE

- [ ] decide whether events, trials or discrete random variables are independent, distinguish pairwise from mutual independence and statistical from linear independence, and compute the probability of at least one event in independent trials

If yesterday carried no information about today, the share of down days after a down day would equal the overall share.

DEFINITION **Independent events**

Events A and B are **independent** if

$$\Pr(A \cap B) = \Pr(A)\Pr(B). \tag{13.6}$$

When $\Pr(A) > 0$, this is equivalent to $\Pr(B \mid A) = \Pr(B)$, by (13.4).

The product form is symmetric and covers $\Pr(A) = 0$. The conditional form says that learning A leaves the probability of B unchanged.

Exhibit 4: Down after down

Yesterday	Down today	Up today	Days	Share down today
Down	1,226	1,417	2,643	0.4639
Up	1,418	1,787	3,205	0.4424
All	2,644	3,204	5,848	0.4521

If independent (a model):

Yesterday	Down today	Up today	Days
Down	1,195.0	1,448.0	2,643
Up	1,449.0	1,756.0	3,205
All	2,644	3,204	5,848

Note: the pair space; down means $R < 0$, and no return is 0, so up is "not down". The second table keeps the margins and puts row total × column total ÷ 5,848 in each cell, the counts that (13.6) would give. Source: as Exhibit 1.

A down day followed a down day 1,226 times, against 1,195.0 under independence with the same margins (Exhibit 4). By (13.6) the sample's signs are not independent, but the gap is 31 days in 5,848. Whether a gap such as this is larger than chance alone produces is a question for the tests of LM21.

Several events need more than pairs. Events $A_1, \ldots, A_m$ are **mutually independent** if, for every choice of two or more of them, the probability that all occur is the product of their probabilities. They are **pairwise independent** if this holds for every pair.

EXAMPLE 5

Pairwise is not mutual

Two independent trades of a fair bet give four equally likely outcomes. Let A_1 be "the first trade wins", A_2 "the second trade wins" and A_3 "the two trades agree". Are the three events independent?

SOLUTION

The events are $A_1 = \{++, +-\}$, $A_2 = \{++, -+\}$ and $A_3 = \{++, --\}$, each with probability 1/2. Each pair meets only in $++$, with probability $1/4 = \frac{1}{2} \cdot \frac{1}{2}$, so the events are pairwise independent. All three also meet only in $++$, so $\Pr(A_1 \cap A_2 \cap A_3) = 1/4$, not $1/8$. Any two of the events fix the third: they are not mutually independent.

LM4 traded its bet "independently of other trades". Formally, trials are **independent** when events about different trials are mutually independent. With win probability p in each of n trials, the product rule gives each sequence with j wins the probability $p^j q^{n-j}$. An event A that each trial produces with the same probability then misses all n trials with probability $(1-\Pr(A))^n$:

$$\begin{aligned} \Pr(\text{a given sequence with } j \text{ wins}) &= p^j q^{n-j}, \\ \Pr(A \text{ at least once in } n \text{ trials}) & \\ = 1 - \big(1 - \Pr(A)\big)^n. & \end{aligned} \tag{13.7}$$

EXAMPLE 6

At least one fall in 20 sessions

A fall of 3% or more has probability $108/5{,}849 = 1.8465\%$ on a TR23 day. If days were independent, what share of blocks of 20 sessions would contain at least one fall? Compare with TR23's consecutive blocks of 20 sessions from 3 January 2003.

SOLUTION

By (13.7), the model gives $1-(1-0.018465)^{20} = 31.1\%$. TR23 has 292 complete blocks, with the last 9 returns excluded, so the model expects about $292 \times 0.311 = 91$ blocks with a fall. The data have 61 such blocks, 20.9%. Falls cluster (Section 3), so several share a block, and the independence model overstates the chance of "at least once".

Statistical independence, a product rule for probabilities, is not the linear independence of vectors in LM9.

PITFALL **Disjoint is not independent**

Disjoint events with positive probabilities are never independent: $\Pr(A \cap B) = 0$, while $\Pr(A)\Pr(B) > 0$. Each excludes the other, a strong dependence. The knowledge check below applies this to up and down days.

DEEP DIVE **Complements of independent events, and counting conditions**

optional · click to expand

Let A and B be independent. Since A is the disjoint union of $A \cap B$ and $A \cap B^c$, we get $\Pr(A \cap B^c) = \Pr(A) - \Pr(A)\Pr(B) = \Pr(A)\Pr(B^c)$. So A and B^c are independent, and the same step applied to B^c and A shows that A^c and B^c are independent too (Grinstead and Snell 1997, §4.1, Exercise 17).

Mutual independence of m events asks for a product rule for every choice of two or more. Of the 2^m sub-collections, m hold one event and one holds none, so $2^m - m - 1$ conditions remain: 4 for three events. Example 5 meets the three pair conditions and fails the fourth (Grinstead and Snell 1997, §4.1, Exercise 7).

導讀 **獨立、互斥與線性獨立**

兩個事件（event）獨立（independent），是指 $\Pr(A \cap B) = \Pr(A)\Pr(B)$，也就是知道 A 發生，不會改變 B 的機率。互斥（disjoint）正好相反：機率都為正的兩個互斥事件，一個發生就排除另一個，所以一定不獨立。這裡的獨立是機率的乘積規則（product rule），和 LM9 向量的線性獨立（linear independence）是兩回事。三個以上的事件，成對獨立（pairwise independence）不代表互相獨立（mutual independence），後者要求每一組事件都滿足乘積規則。TR23 的大跌（跌 3% 以上）會聚集出現，所以「每天獨立」的模型算出「20 個交易日內至少一次大跌」的機率是 31.1%，高估了實際的 20.9%。

KNOWLEDGE CHECK 3

On TR23, $\Pr(\text{up}) = 3{,}205/5{,}849 = 0.5480$ and $\Pr(\text{down}) = 2{,}644/5{,}849 = 0.4520$. Are "up today" and "down today" independent?

ANSWER

No. A day cannot be up and down at once, so the events are disjoint. The probability of both is 0, while the product is $0.5480 \times 0.4520 = 0.2477$. Each excludes the other.

5 TOTAL PROBABILITY AND BAYES' RULE

- ☐ apply the law of total probability and Bayes' rule with base rates, counts and odds, and explain why a conditional probability and its inverse can differ several times over

A **partition** of Ω is a list of disjoint events $A_1, \ldots, A_m$ whose union is Ω.

THEOREM **Law of total probability**

Let $A_1, \ldots, A_m$ be a partition of Ω with $\Pr(A_i) > 0$ for each i. Then for every event B,

$$\Pr(B) = \sum_{i=1}^{m} \Pr(B \mid A_i) \Pr(A_i). \tag{13.8}$$

Proof. The events $B \cap A_i$ are disjoint with union B, so additivity gives $\Pr(B) = \sum_i \Pr(B \cap A_i)$. The multiplication rule (13.5) writes each term as $\Pr(B \mid A_i) \Pr(A_i)$. ■

TR23 splits at 1 June 2015, when the daily price limit was raised from 7% to 10% (LM5). Exhibit 5 compares the two regimes.

Exhibit 5: Two limit regimes

	7% limit	10% limit	All
Days	3,077	2,772	5,849
Share of days	52.61%	47.39%	100%
Falls of 3% or more	72	36	108
Share of days that fell	2.34%	1.30%	1.85%
After a big move: falls / days	17 / 303	15 / 186	32 / 489
Share that fell after a big move	5.61%	8.06%	6.54%
Hit rate: share of falls after a big move	23.61%	41.67%	29.63%

Note: returns dated 3 January 2003 → 29 May 2015 and 1 June 2015 → 8 October 2026. The last three rows use the 3,076 and 2,771 day pairs inside one regime. The pair of 29 May and 1 June 2015 (−0.12%, then −0.77%) lies in neither and changes no count. A stability check. Source: as Exhibit 1.

EXAMPLE 7

Falls across the two limit regimes

Recombine the fall rates of Exhibit 5 into TR23's with (13.8), and find the probability that a fall of 3% or more came from the 10% regime.

SOLUTION

Let A be a fall of 3% or more. The two regimes partition the 5,849 days, so by (13.8), in counts,

$$\begin{aligned}\Pr(A) &= \frac{72}{3{,}077}\cdot\frac{3{,}077}{5{,}849} + \frac{36}{2{,}772}\cdot\frac{2{,}772}{5{,}849}\\ &= \frac{108}{5{,}849} = 1.85\%\,:\end{aligned}$$

the fall rates 2.34% and 1.30%, weighted by the shares of days, 52.6% and 47.4%.

By (13.4), a fall came from the 10% regime with probability

$$\frac{36/5{,}849}{108/5{,}849} = 36/108 = 33.3\%,$$

against the regime's 47.4% of days. Rebuilding the denominator from the rounded 2.34%, 1.30%, 52.6% and 47.4% gives 33.4%, so compute from the counts.

Example 7 reversed a conditional probability, from the fall rate of a regime to the regime of a fall. Applying (13.4) twice does this in general: $\Pr(A_i \cap B)$ equals both $\Pr(A_i \mid B)\Pr(B)$ and $\Pr(B \mid A_i)\Pr(A_i)$.

THEOREM **Bayes' rule**

Let $A_1, \ldots, A_m$ be a partition of Ω with each $\Pr(A_i) > 0$, and let $\Pr(B) > 0$. Then

$$\Pr(A_i \mid B) = \frac{\Pr(B \mid A_i)\Pr(A_i)}{\Pr(B)}, \tag{13.9}$$

with $\Pr(B)$ from (13.8). For the partition A, A^c the denominator is

$$\begin{aligned}\Pr(B) = {}&\Pr(B \mid A)\Pr(A)\\ &+ \Pr(B \mid A^c)\Pr(A^c).\end{aligned}$$

In the two-event form, $\Pr(A)$ is the **base rate** of A, $\Pr(B \mid A)$ the **hit rate** of the signal B, and $\Pr(B \mid A^c)$ its **false-alarm rate**. Before B is observed, $\Pr(A)$ is the **prior probability** of A; after it, $\Pr(A \mid B)$ is the **posterior probability**.

Dividing (13.9) for A by the same formula for A^c cancels $\Pr(B)$. With the **odds** of A defined as $\Pr(A)/\Pr(A^c)$,

$$\frac{\Pr(A \mid B)}{\Pr(A^c \mid B)} = \frac{\Pr(A)}{\Pr(A^c)} \times \frac{\Pr(B \mid A)}{\Pr(B \mid A^c)}. \tag{13.10}$$

Posterior odds equal prior odds times the **likelihood ratio**, the hit rate divided by the false-alarm rate. The form needs $0 < \Pr(A) < 1$ and $\Pr(B \mid A^c) > 0$. Odds convert back to a probability by dividing them by one plus the odds.

EXAMPLE 8

Resolving the case: Bayes' rule three ways

In the case, A is "a fall of 3% or more today" and B "a move of 2% or more either way yesterday", on the pair space. Find $\Pr(A \mid B)$ from Exhibit 1 in three ways.

SOLUTION

(a) Read Exhibit 1 down its columns, as a tree of counts: 32 of the 108 falls and 457 of the 5,740 other days followed a big move. Of the $32 + 457 = 489$ days after a big move, 32 fell: $32/489 = 6.54\%$.

(b) With four significant digits, $\Pr(A) = 1.847\%$, $\Pr(B \mid A) = 29.63\%$ and $\Pr(B \mid A^c) = 7.962\%$, the two terms of $\Pr(B)$ and the ratio (13.9) are

$$\begin{aligned} 0.01847 \times 0.2963 &= 0.005473, \\ 0.98153 \times 0.07962 &= 0.07815, \\ \frac{0.005473}{0.005473 + 0.07815} &= 6.54\%. \end{aligned}$$

The case's rounded inputs, 1.85%, 29.6% and 7.96%, would give 6.55%: carry four digits, or work from counts.

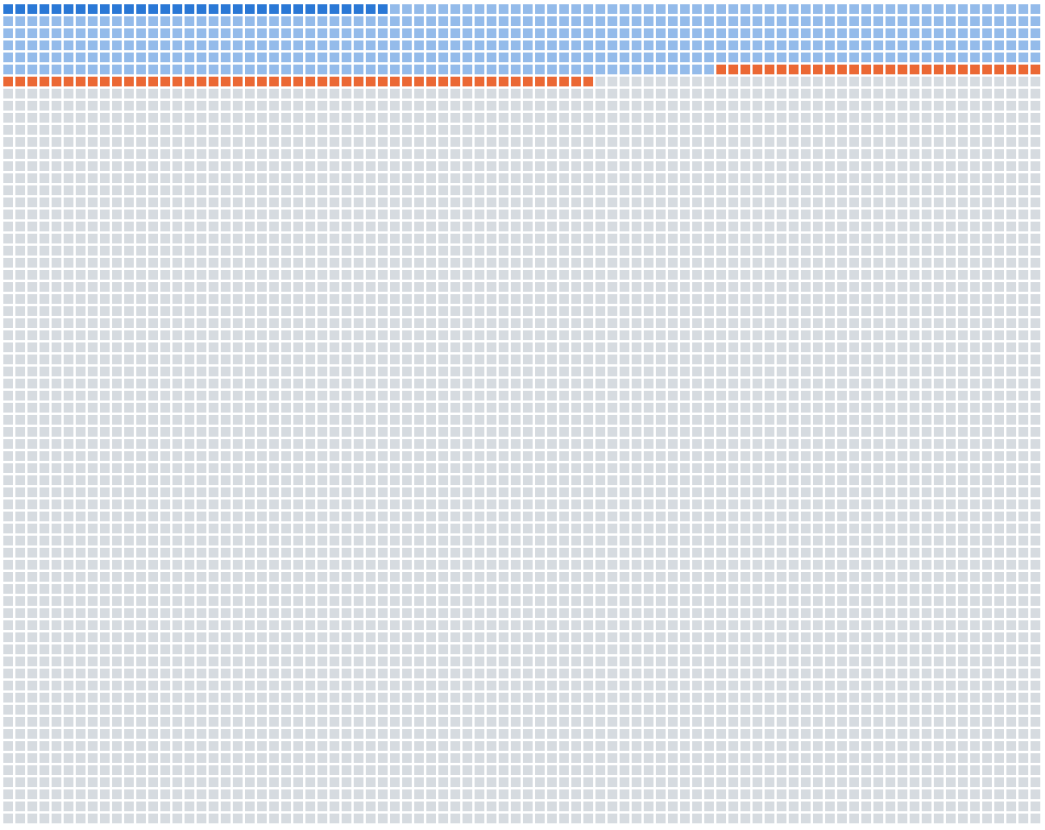
(c) By (13.10), the prior odds are $108/5{,}740 = 0.018815$, and the counts give the likelihood ratio $(32/108)/(457/5{,}740) = 3.7215$. The posterior odds are $0.018815 \times 3.7215 = 0.07002$, which is $32/457$, and the probability is $0.07002/1.07002 = 6.54\%$.

So 29.63% and 6.54% answer different questions. The base rate is only 1.85%, and the false alarms on 7.96% of 5,740 days outnumber the hits, 457 to 32.

Counts help. In Study 1 of Gigerenzer and Hoffrage (1995), data stated as frequencies rather than probabilities raised the share of Bayesian answers. It rose from 16% to 46% with their standard menu and from 28% to 50% with the short one. The widget below draws the counts, one square per day.

From a hit rate to a conditional probability

INTERACTIVE



Fall after a big move: 32 | Big move, no fall: 457 | Fall after a smaller move: 76 | Neither: 5,283

Pr(B): big move yesterday
8.36%

Pr(A | B): fall after a big move
6.54%

Pr(A | not B): fall after a smaller move
1.42%

Pr(A | B) ÷ 108/5,848
3.54

Likelihood ratio, hit ÷ false alarm
3.7215

Prior odds × ratio = posterior odds
0.018815 × 3.7215 = 0.07002

In the picture: falls among the big-move days
32 of 489, 6.5%

Preset: TR23's exact rates 108/5,848, 32/108 and 457/5,740 on the pair space (descriptive). The squares are the sample's days. A readout whose denominator is 0 is undefined.

EXAMPLE 9

A manager who beat the index

A fund manager beat the index in 8 of 12 months, a hypothetical record. Before the record, 10% of managers are skilled. A skilled manager beats the index in a month with probability 0.55, any other with 0.5. Given the manager's type, months are independent. Find the probability that this manager is skilled.

SOLUTION

By (13.7), applied within each type, a given sequence of 8 good and 4 bad months has probability $0.55^8 \times 0.45^4$ for a skilled manager and 0.5^{12} for the others. The likelihood ratio is

$$\frac{0.55^8 \times 0.45^4}{0.5^{12}} = 1.4064.$$

By (13.10), the posterior odds are $(0.1/0.9) \times 1.4064 = 0.15627$, so $\Pr(\text{skilled} \mid \text{record}) = 0.15627/1.15627 = 13.5\%$. Eight good months in twelve move a 10% prior only to 13.5%.

Counting the orders in which the good months can fall leads to the binomial distribution of LM15; Bayes' rule does not need that count, because it cancels. LM24 applies the same rule to an unknown parameter.

PITFALL **The prosecutor's fallacy**

Reading $\Pr(B \mid A)$ as $\Pr(A \mid B)$ is the prosecutor's fallacy. The probability of the evidence given innocence is not the probability of innocence given the evidence. "29.6% of falls follow a big move" does not mean "29.6% of big moves are followed by a fall"; that share is 6.54%. The two differ by the factor $\Pr(B)/\Pr(A) = 489/108 = 4.53$.

導讀 **全機率法則與貝氏定理**

全機率法則（law of total probability）把樣本空間（sample space）切成互斥（disjoint）的幾塊，分塊計算後再加權加總：漲跌幅限制 7% 與 10% 兩個時期的大跌（跌 3% 以上）比例 2.34% 與 1.30%，按天數比重加權，就回到全樣本的 1.85%。貝氏定理（Bayes' rule）把條件反過來：基本率（base rate）$\Pr(A)$ 是事前機率（prior probability），看到訊號 B 之後的 $\Pr(A \mid B)$ 是事後機率（posterior probability）。前一天漲跌 2% 以上之後隔天大跌，489 天中只有 32 天，是 6.54%；但 108 次大跌中有 32 次發生在前一天漲跌 2% 以上之後，是 29.6%。把這兩個比例混為一談，就是檢察官謬誤（prosecutor's fallacy）。用 odds 計算最簡單：事後 odds 等於事前 odds 乘上 likelihood ratio。

KNOWLEDGE CHECK 4

With the case's hit and false-alarm rates, a likelihood ratio of 3.7215, what base rate would make $\Pr(A \mid B) = 50\%$?

ANSWER

Posterior odds of 1 need prior odds of $1/3.7215$, by (13.10). These odds belong to the probability $1/(1 + 3.7215) = 21.18\%$, more than eleven times TR23's base rate of 1.85%.

6 RANDOM VARIABLES AND THEIR DISTRIBUTIONS

- [ ] decide whether events, trials or discrete random variables are independent, distinguish pairwise from mutual independence and statistical from linear independence, and compute the probability of at least one event in independent trials

- [ ] define a random variable as a function on the sample space, and find the probability mass function and distribution function of a discrete random variable, an indicator included

A trader cares less about the outcome $+-$ than about the wealth it leaves.

EXAMPLE 10

A random variable on two trades

On the sample space of Example 1, at $f = 25$, let $X = W_2/W_0$ be wealth after two trades as a multiple of the start. Tabulate the values of X, their probabilities and the per-trade log growth $\frac{1}{2}\ln X$.

SOLUTION

Each outcome gives one value: $X(++) = 1.5^2 = 2.25$, $X(+-) = X(-+) = 1.5 \times 0.75 = 1.125$ and $X(--) = 0.75^2 = 0.5625$. Two outcomes share the value 1.125, so $\Pr(X = 1.125) = 1/2$, while 2.25 and 0.5625 have probability 1/4 each. Wealth ends below its start on the event $\{X < 1\} = \{--\}$, with probability 1/4.

The per-trade log growth $\frac{1}{2}\ln X$, a function of X, is a random variable too. Its values are $\ln 1.5 = 40.55\%$, $\frac{1}{2}\ln 1.125 = 5.889\%$ and $\ln 0.75 = -28.77\%$; the middle one is the g^* of LM4. Since $\frac{1}{2}(\ln 1.5 + \ln 0.75) = g^*$, the probability-weighted average is exactly g^*:

$$\begin{aligned} \tfrac{1}{4}\ln 1.5 + \tfrac{1}{2}g^* + \tfrac{1}{4}\ln 0.75 &= \tfrac{1}{2}g^* + \tfrac{1}{2}g^* \\ &= g^*. \end{aligned}$$

The probability-weighted average of a random variable is its expectation; LM14 defines it and connects it with (5.9) of LM5.

DEFINITION **Random variable, probability mass function, distribution function**

A **random variable** is a function $X : \Omega \to \mathbb{R}$ that assigns a number to each outcome. For a number x, $\{X \le x\}$ is the event $\{\omega : X(\omega) \le x\}$. A random variable is **discrete** if it takes finitely or countably many values; its **probability mass function** (pmf) is $p_X(x) := \Pr(X = x)$. Every random variable has a **distribution function** F_X, which gives the probability of an interval:

$$\begin{aligned} F_X(x) &:= \Pr(X \le x), \\ \Pr(x_{\text{lo}} < X \le x_{\text{hi}}) &= F_X(x_{\text{hi}}) - F_X(x_{\text{lo}}). \end{aligned} \tag{13.11}$$

The interval rule holds because, for $x_{\text{lo}} < x_{\text{hi}}$, the event $\{X \le x_{\text{hi}}\}$ is the disjoint union of $\{X \le x_{\text{lo}}\}$ and $\{x_{\text{lo}} < X \le x_{\text{hi}}\}$. These are the conventions of LM5, where a density gave the distribution function; here F_X jumps.

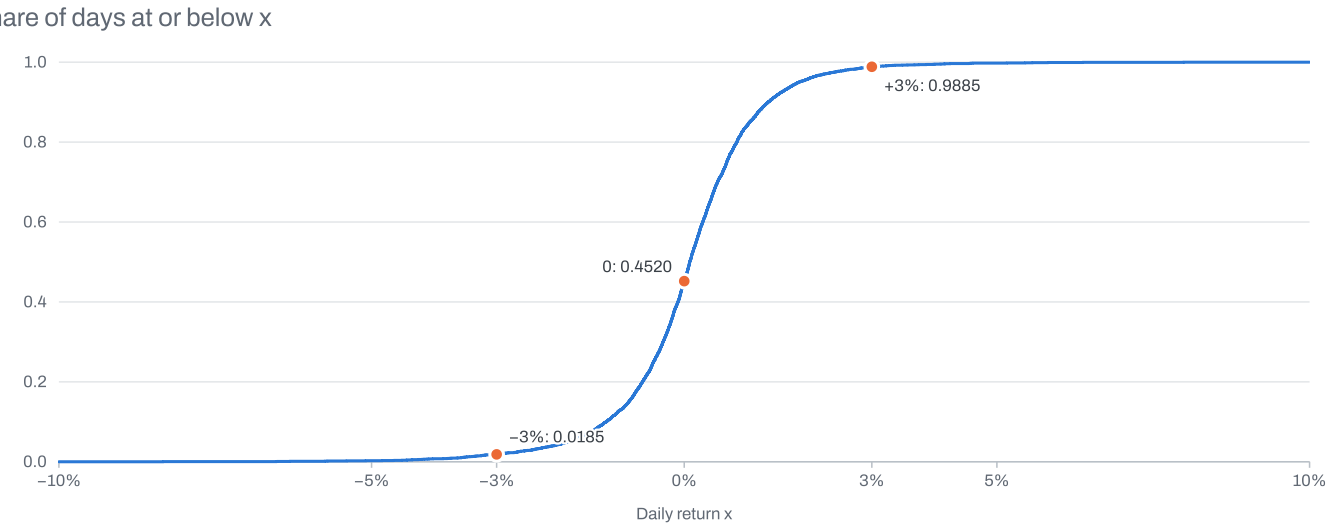
The **indicator** of an event A is the random variable $\mathbf{1}_A$, equal to 1 on A and 0 elsewhere; $\mathbf{1}\{\cdot\}$ names the event inside the braces. In Example 10, $\mathbf{1}\{X < 1\}$ is 1 only on $--$, so its pmf is 3/4 at 0 and 1/4 at 1.

On TR23, the return R of a day drawn at random is a random variable of the empirical model. Its distribution function is the **empirical distribution function**:

$$\hat{F}(x) = \frac{\#\{t : R_t \le x\}}{T}. \tag{13.12}$$

Exhibit 6 draws $\hat{F}$: the function rises by $1/5{,}849$ at each return. It is 0.0185 at −3%, the 108 falls; 0.4520 at 0; and 0.9885 at +3%.

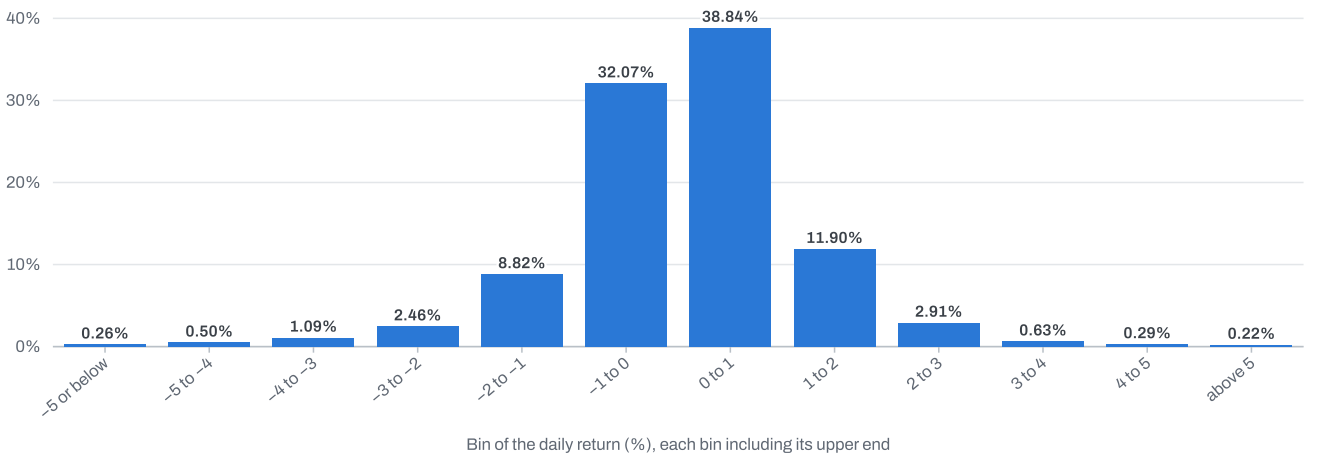
Exhibit 6: The distribution function of a TAIEX day



Note: $\hat{F}(x)$, the share of TR23's 5,849 daily returns at or below x, a step function that jumps by $1/5{,}849$ at each return. The smallest return, −9.70% on 7 April 2025, and the largest, +9.25% on 10 April 2025, lie inside the axis. On 7 April 2025 many stocks closed limit-down, which capped that day's fall. Source: as Exhibit 1.

A function of a random variable is a random variable. Placing each return in its 1% bin gives a discrete one, whose pmf Exhibit 7 shows; the first three bars hold the 108 falls, $\hat{F}(-3\%)$.

Exhibit 7: A function of R: the 1% bin



Note: the pmf of the bin of a TR23 day, 5,849 returns, in bins that hold their upper ends, with the tails beyond ±5% pooled. Counts from the left: 15, 29, 64, 144, 516, 1,876, 2,272, 696, 170, 37, 17 and 13. The printed shares are rounded and sum to 99.99%. Source: as Exhibit 1.

The product rule extends from events to random variables.

DEFINITION **Independent random variables**

Discrete random variables $X_1, \ldots, X_n$ are **independent** if, for all values $x_1, \ldots, x_n$,

$$\begin{aligned} &\Pr(X_1 = x_1, \ldots, X_n = x_n) \\ &\quad = p_{X_1}(x_1) \cdots p_{X_n}(x_n). \end{aligned} \tag{13.13}$$

Independent random variables that share one pmf are **independent and identically distributed** (iid).

Functions of independent random variables, one for each, are independent too: summing (13.13) over the values that each function merges keeps the product form. The win indicators of n independent trades of LM4's bet are iid, each equal to 1 with probability p. For random variables with densities, such as the normal draws of LM12, LM16 states independence through their joint distribution.

PITFALL **The distribution function counts the point itself**

By (13.11), $F_X(x) = \Pr(X \le x)$ includes the value x. In Example 10, $F_X(1.125) = 3/4$, while $\Pr(X < 1.125) = 1/4$; the difference is the jump $p_X(1.125) = 1/2$. Strict and inclusive inequalities differ at every value with positive probability.

DEEP DIVE **Properties of a distribution function** optional · click to expand

Every distribution function F_X is non-decreasing: for $x_{\text{lo}} < x_{\text{hi}}$, the event $\{X \le x_{\text{lo}}\}$ lies inside $\{X \le x_{\text{hi}}\}$. It is right-continuous: as x_{hi} decreases to x_{lo}, the events $\{X \le x_{\text{hi}}\}$ decrease to $\{X \le x_{\text{lo}}\}$, and the continuity of Deep Dive 1 applies. The events also decrease to $\emptyset$ as $x \to -\infty$ and increase to Ω as $x \to \infty$, so F_X tends to 0 and to 1.

From the left, the events $\{X \le x\}$ increase to $\{X < x_{\text{hi}}\}$ as x increases to x_{hi}. So the jump of F_X at x_{hi} is $\Pr(X = x_{\text{hi}})$, and F_X is continuous where $\Pr(X = x) = 0$.

導讀 **隨機變數是函數**

隨機變數（random variable）不是「隨機的數」，而是把樣本空間（sample space）裡每一個結果（outcome）對應到一個數字的函數：兩筆交易的四個結果，對應到財富倍數 2.25、1.125、1.125 與 0.5625。離散型（discrete）隨機變數用機率質量函數（probability mass function）p_X 描述；任何隨機變數都有累積分布函數（distribution function）$F_X(x) = \Pr(X \le x)$，它把 x 本身也算進去，所以在跳躍點上 $\Pr(X < x)$ 和 $F_X(x)$ 不同。指示函數（indicator）$\mathbf{1}_A$ 在事件（event）A 發生時為 1，否則為 0。幾個離散隨機變數的聯合機率（joint probability）若等於各自機率的乘積，它們就是獨立的（independent）；若分布（distribution）也相同，就是獨立同分布（independent and identically distributed, iid）。

KNOWLEDGE CHECK 5

Is $\hat{F}(0)$ the share of TR23's down days? What is $\Pr(-1\% < R \le 1\%)$?

ANSWER

Yes on TR23: $\hat{F}(0) = 2{,}644/5{,}849 = 0.4520$ counts the days with $R \le 0$, and no return equals 0. In general the two differ by $\Pr(R = 0)$. By (13.11),

$$\begin{aligned}\Pr(-1\% < R \le 1\%) &= \hat{F}(1\%) - \hat{F}(-1\%) \\ &= (4{,}916 - 768)/5{,}849 \\ &= 4{,}148/5{,}849 = 0.7092.\end{aligned}$$

7 PROBABILITY IN PYTHON

- ☐ compute conditional frequencies, Bayes' rule and independence checks from a return series in Python, counting threshold days with a tolerance for exact ties

Counting events in Polars takes three steps: yesterday's return on today's row, events as booleans with a tie guard, and exact arithmetic to confirm the guard.

PYTHON

```python
from fractions import Fraction
import polars as pl

EPS = 1e-12                    # guard for exact ties at a threshold
tr = pl.read_csv("data/taiex_tr.csv").sort("date")   # date, price
df = tr.with_columns(R=pl.col("price") / pl.col("price").shift(1) - 1)
df = df.with_columns(R_prev=pl.col("R").shift(1))    # yesterday's R
pairs = df.drop_nulls("R_prev")                      # 5,848 days
fall = pl.col("R") <= -0.03 + EPS                    # A, today
big = pl.col("R_prev").abs() >= 0.02 - EPS           # B, yesterday
c = pairs.select(AB=(fall & big).sum(), A=fall.sum(), B=big.sum(),
                 N=pl.len()).row(0, named=True)
print(c)                     # {'AB': 32, 'A': 108, 'B': 489, 'N': 5848}
base, hit = c["A"] / c["N"], c["AB"] / c["A"]
alarm = (c["B"] - c["AB"]) / (c["N"] - c["A"])
post = base * hit / (base * hit + (1 - base) * alarm)   # (13.9)
print(f"{post:.6f} {c['AB'] / c['B']:.6f}")            # 0.065440 0.065440

# exact check: no return equals -3% (the closes have two decimals)
closes = [Fraction(f"{x:.2f}") for x in tr["price"]]
rets = [b / a - 1 for a, b in zip(closes, closes[1:])]
print(sum(r == Fraction(-3, 100) for r in rets))        # 0
```

The method `shift(1)` moves each value one row down and leaves a null in the first row (Polars reference). Dropping the nulls of `R_prev` keeps the 5,848 days with a yesterday. The `Fraction` recount from the two-decimal closes finds no return equal to 0, ±1%, ±2%, ±3%, ±4% or ±5%, and 5,849 distinct returns.

> **PITFALL** **Exact ties at a threshold**
>
> Two-decimal prices make round returns exact: 176.00 → 158.40 is −10% exactly. Floating-point division gives −0.09999999999999998, which `R <= -0.10` misses. Count every threshold with one guard, `<= x + 1e-12` and `>= x - 1e-12`, and confirm it with exact fractions. TR23 has no such tie, but the guard costs nothing.

IN PYTHON **Companion lab**

The Polars-first notebook `LM13_lab.ipynb` rebuilds the module's exhibits and computational examples from the TR23 closes, recounts the threshold days with exact fractions and solves the computational practice problems. It ends with two extensions: the fall share after moves of several sizes in each regime, and the share of falls with a big move in the previous five sessions.

The notebook reads FinMind files that you download once with your own token. From the repository root, run `uv run python data/fetch_finmind.py`, then `uv run python data/make_lab_data.py`. FinMind's licence does not allow the book to redistribute the data. After that, the notebook runs offline.

KNOWLEDGE CHECK 6

What does `shift(1)` put in the first row, and why does the pair space have 5,848 days?

ANSWER

A null, because the first row has no row before it. The column `R` already has a null in its first row, since the first close has no previous close. So `R_prev` has nulls in its first two rows, and dropping them leaves $5{,}850 - 2 = 5{,}848$ days.

SUMMARY

- Three axioms define a probability on the events of a sample space; the complement rule, monotonicity, (13.1) and (13.2) follow.
- Equally likely outcomes give (13.3). Under the empirical distribution, every frequency of TR23 is an exact probability.
- Conditioning (13.4) shrinks the sample space, and (13.5) chains it: 32 of the 489 days after a big move fell 3% or more, 6.54% against 1.85%.
- Independence is (13.6). Signs of consecutive days come close to it; falls cluster. Pairwise is not mutual independence, and disjoint events with positive probabilities are dependent.
- For independent trials, (13.7) gives 31.1% for a fall in 20 sessions, against 20.9% of TR23's blocks.
- Total probability (13.8), Bayes' rule (13.9) and the odds form (13.10) turn 1.847%, 29.63% and 7.962% into 6.54%. Reading one direction as the other is the prosecutor's fallacy.
- A random variable is a function on the sample space, with a pmf and a distribution function (13.11); TR23 has $\hat{F}$ (13.12). Discrete random variables are independent when their joint pmf factors (13.13).

PRACTICE PROBLEMS

1. On TR23's 5,849 returns, with inclusive thresholds, $\Pr(R \le -2\%) = 4.31\%$ and $\Pr(R \ge 2\%) = 4.05\%$. Which value is *closest* to $\Pr(|R| \ge 2\%)$?

   A. 0.17%

   B. 8.19%

   C. 8.36%

2. On the 2,771 day pairs with both days in the 10% limit regime, a fall of 3% or more has a base rate of 1.299%. A big move yesterday has a hit rate of 41.67% and a false-alarm rate of 6.252%. Which value is *closest* to the probability of a fall after a big move?

   A. 41.7%

   B. 8.06%

   C. 1.30%

3. In Example 5, replace A_3 by "the two trades disagree". The events A_1, A_2 and the new A_3 are:

   A. pairwise but not mutually independent.

   B. mutually independent.

   C. not even pairwise independent.

4. Derive inclusion–exclusion for three events from (13.1) applied three times. Apply it to TR23's 5,847 triples of consecutive days, with A_i the triples whose day i fell 2% or more. Each A_i holds 252 triples; A_1 and A_2 share 38, A_1 and A_3 share 34, A_2 and A_3 share 38, and all three share 8. Compare the union with (13.2), in percent to two decimals.

5. Of TR23's 5,847 triples, 2,642 have a down first day ($R < 0$), 1,225 a down first and second day, and 542 three down days. Days 1, 2 and 3 are down in 2,642, 2,643 and 2,644 triples. Use (13.5), with factors to four decimals, for three down days in a row; compare with independent days with these daily shares, and comment on the third factor.

6. A fall of 3% or more has probability $108/5{,}849$ on a TR23 day. Find the smallest number n of independent sessions for which (13.7) gives at least one fall with probability 50% or more. Of TR23's consecutive blocks of n sessions from 3 January 2003, the incomplete last block excluded, 50 hold a fall. Explain the gap.

7. Repeat Example 7 for falls of 5% or more: 8 of the 3,077 days of the 7% regime and 7 of the 2,772 days of the 10% regime. Give both shares in percent to four decimals and the likelihood ratio of the 10% regime to three decimals. What is the probability that such a fall came from the 10% regime?

8. Repeat Example 9 for a record of 16 good months in 24. Then find the prior share of skilled managers that would make the record of 8 good months in 12 give a posterior of 50%. Give likelihood ratios to four decimals and probabilities in percent to two decimals.

9. Trade LM4's +2%/−1% bet three times at $f = 25$, independently, with $p = 1/2$. List the outcomes, find the pmf of $X = W_3/W_0$ and $\Pr(W_3 < W_0)$. Why can this probability be so large when $g^* > 0$?

10. On the pair space, let $X = \mathbf{1}\{R_{t-1} < 0\}$ and $Y = \mathbf{1}\{R_t < 0\}$. Use Exhibit 4 to show, to four decimals, that X and Y are not independent in the empirical model. Does this alone show that the sign of one day depends on the day before beyond these 5,848 days?

11. *(Python)* From `taiex_tr.csv`, rebuild Exhibit 1 with `shift(1)` and the guard. Repeat it on the day pairs with both days in the 10% regime, and confirm with `Fraction` that no return equals ±2% or −3%.

12. Two independent trades each win with probability 0.55, and X is the number of losing trades. Which value is *closest* to $F_X(1)$?

    A. 0.4950

    B. 0.6975

    C. 0.7975

SOLUTIONS

1. **C** is correct. A day cannot fall 2% and rise 2%, so the events are disjoint and their probabilities add: $4.31\% + 4.05\% = 8.36\%$, exactly $489/5{,}849$. A is the product, which reads "or" as "and". B is $1 - (1 - 0.0431)(1 - 0.0405) = 8.19\%$, which subtracts an overlap that disjoint events lack.

2. **B** is correct. The two terms of the denominator of (13.9), and the ratio, are

$$\begin{aligned} 0.01299 \times 0.4167 &= 0.005413, \\ 0.98701 \times 0.06252 &= 0.06171, \\ \frac{0.005413}{0.005413 + 0.06171} &= 8.06\%, \end{aligned}$$

   exactly $15/186$ as in Exhibit 5. A is the hit rate, the inverted conditional probability. C is the base rate, which ignores the signal.

3. **A** is correct. The new event is $A_3 = \{+-, -+\}$, with probability 1/2. The pairs meet in one outcome each, $A_1 \cap A_2 = \{++\}$, $A_1 \cap A_3 = \{+-\}$ and $A_2 \cap A_3 = \{-+\}$, of probability $1/4 = \frac{1}{2} \cdot \frac{1}{2}$, so every pair is independent. No outcome lies in all three events, so the triple has probability $0 \neq 1/8$. B takes the triple product for granted without counting. C fails: A_3 is built from A_1 and A_2, yet $\Pr(A_1 \cap A_3) = \Pr(\{+-\}) = 1/4$ is the product.

4. Apply (13.1) to $A_1 \cup A_2$ and A_3, then to A_1 and A_2, and then to $A_1 \cap A_3$ and $A_2 \cap A_3$, whose union is $(A_1 \cup A_2) \cap A_3$:

$$\begin{aligned}
&\Pr(A_1 \cup A_2 \cup A_3) \\
&\quad = \Pr(A_1 \cup A_2) + \Pr(A_3) \\
&\qquad - \big[\Pr(A_1 \cap A_3) + \Pr(A_2 \cap A_3) \\
&\qquad\quad - \Pr(A_1 \cap A_2 \cap A_3)\big] \\
&\quad = \Pr(A_1) + \Pr(A_2) + \Pr(A_3) \\
&\qquad - \Pr(A_1 \cap A_2) - \Pr(A_1 \cap A_3) \\
&\qquad - \Pr(A_2 \cap A_3) + \Pr(A_1 \cap A_2 \cap A_3).
\end{aligned}$$

   The union holds $3 \times 252 - 38 - 34 - 38 + 8 = 654$ triples, 11.19%. The union bound gives $756/5{,}847 = 12.93\%$, too high by the triples it counts more than once.

5. The chain gives $0.4519 \times 0.4637 \times 0.4424 = 9.27\%$, from $2{,}642/5{,}847$, $1{,}225/2{,}642$ and $542/1{,}225$, and $542/5{,}847$ gives the same 9.27%. Independent days with the shares 0.4519, 0.4520 and 0.4522 give 9.24%. The second factor, 0.4637, exceeds the daily share, but the third, 0.4424, falls below it. The chain itself needs no independence.

6. Solve $1 - (1 - 108/5{,}849)^n \geq 0.5$: $n \geq \ln 0.5 / \ln(1 - 108/5{,}849) = 37.19$, so $n = 38$, with 50.75% at 38 against 49.82% at 37. TR23 has 153 blocks of 38 sessions, the last 35 returns excluded, and 50 of them hold a fall, 32.68%. Falls cluster (Section 3), so several land in one block, and fewer blocks hold one than independent days would give.

7. The shares are $8/3{,}077 = 0.2600\%$ and $7/2{,}772 = 0.2525\%$, so the likelihood ratio is $0.2525/0.2600 = 0.971$. By (13.10), the posterior odds are $(2{,}772/3{,}077) \times 0.971$, and the probability is $7/15 = 46.7\%$, against the prior 47.4%. A ratio near 1 hardly moves a probability.

8. The likelihood ratio is $0.55^{16} \times 0.45^8 / 0.5^{24} = 1.9780$, the square of 1.4064, because the record is the old one twice. The posterior odds are $(0.1/0.9) \times 1.9780 = 0.21978$, a posterior of 18.02%. A posterior of 50% after 8 of 12 needs prior odds of $1/1.4064$, a prior of $1/(1 + 1.4064) = 41.56\%$.

9. The outcomes are $+++$, $++-$, $+-+$, $-++$, $+--$, $-+-$, $--+$ and $---$, each with probability 1/8 by (13.7). The multiple X depends only on the number of wins: $1.5^3 = 3.375$ (1/8), $1.5^2 \times 0.75 = 1.6875$ (3/8), $1.5 \times 0.75^2 = 0.84375$ (3/8) and $0.75^3 = 0.421875$ (1/8). So $\Pr(W_3 < W_0) = 1/2$. A positive g^* describes the long run of LM17; over three trades, one win and two losses already leave wealth below its start.

10. In the empirical model, the joint probability and the product of the margins are

$$\begin{aligned}
&\Pr(X = 1, Y = 1) \\
&\quad = 1{,}226/5{,}848 = 0.2096, \\
&\Pr(X = 1)\Pr(Y = 1) \\
&\quad = 0.4519 \times 0.4521 = 0.2043,
\end{aligned}$$

   so (13.13) fails. Within the empirical model the signs are dependent, and so, by the rule for functions of independent random variables (Section 6), are the returns. Beyond these days, no: that

model describes these 5,848 days only. Even independent signs would show some gap in a finite sample, and LM21 asks whether this one is larger than chance alone produces.

11. The rebuild gives 32, 76, 457 and 5,283 days, as in Exhibit 1. On the 2,771 day pairs of the 10% regime, the same table reads:

Yesterday's move	Fall of 3% or more today	Other days	All days	Share that fell
2% or more either way	15	171	186	8.06%
Smaller	21	2,564	2,585	0.81%
All	36	2,735	2,771	1.30%
Share after a big move	41.67%	6.25%	6.71%	

So 15 of the 186 days after a big move fell 3% or more, 8.06%, and the 15 are 41.67% of the regime's 36 falls, as in Exhibit 5. No return equals ±2% or −3%, so the guard changes no count.

12. **C** is correct. With $p = 0.55$ and $q = 0.45$, the pmf of X is p^2, $2pq$ and q^2 at 0, 1 and 2 losses, so $F_X(1) = 1 - q^2 = 0.7975$. A, $2pq = 0.4950$, is the pmf at 1, not the distribution function. B, $1 - p^2 = 0.6975$, counts wins instead of losses.

GLOSSARY

Term	中文	Meaning
Sample space	樣本空間	The set Ω of all outcomes of a random experiment.
Outcome	結果	One element ω of Ω.
Event	事件	A subset of Ω; it occurs when the outcome lies in it.
Complement	餘事件	A^c, the outcomes not in A; $\Pr(A^c) = 1 - \Pr(A)$.
Disjoint events	互斥事件	Events with no outcome in common, $A \cap B = \emptyset$.
Axioms of probability	機率公設	$\Pr(A) \geq 0$, $\Pr(\Omega) = 1$ and additivity over disjoint events.
Inclusion–exclusion	沿用英文	$\Pr(A \cup B) = \Pr(A) + \Pr(B) - \Pr(A \cap B)$, (13.1).
Union bound	沿用英文	The probability of a union is at most the sum of the probabilities, (13.2).
Empirical distribution	經驗分布	Probability $1/T$ on each of the T observations of a sample.
Conditional probability	條件機率	$\Pr(B \mid A) = \Pr(A \cap B) / \Pr(A)$ for $\Pr(A) > 0$, (13.4).
Multiplication rule	乘法規則	$\Pr(A \cap B) = \Pr(A) \Pr(B \mid A)$ and its chain, (13.5).
Independent events	獨立事件	Events with $\Pr(A \cap B) = \Pr(A) \Pr(B)$, (13.6).
Pairwise independence	成對獨立	The product rule for every pair of events.
Mutual independence	互相獨立	The product rule for every choice of two or more events.
Partition	沿用英文	Disjoint events whose union is Ω.
Law of total probability	全機率法則	$\Pr(B) = \sum_i \Pr(B \mid A_i) \Pr(A_i)$ over a partition, (13.8).
Bayes' rule	貝氏定理	$\Pr(A_i \mid B) = \Pr(B \mid A_i) \Pr(A_i) / \Pr(B)$, (13.9).
Base rate	基本率	$\Pr(A)$, the probability of A before a signal is observed.

Term	中文	Meaning
Prior and posterior probability	事前機率、事後機率	$\Pr(A)$ before and $\Pr(A \mid B)$ after observing B.
Hit rate, false-alarm rate	沿用英文	$\Pr(B \mid A)$ and $\Pr(B \mid A^c)$ for a signal B.
Odds, likelihood ratio	沿用英文	$\Pr(A)/\Pr(A^c)$; $\Pr(B \mid A)/\Pr(B \mid A^c)$, (13.10).
Prosecutor's fallacy	檢察官謬誤	Reading $\Pr(B \mid A)$ as $\Pr(A \mid B)$.
Random variable	隨機變數	A function from Ω to the real numbers.
Probability mass function	機率質量函數	$p_X(x) = \Pr(X = x)$ for a discrete random variable.
Distribution function	累積分布函數	$F_X(x) = \Pr(X \le x)$, (13.11).
Empirical distribution function	經驗分布函數	$\hat{F}(x)$, the share of a sample at or below x, (13.12).
Indicator	指示函數	$\mathbf{1}_A$, equal to 1 on A and 0 elsewhere.
iid	獨立同分布	Independent random variables that share one distribution, (13.13).

REFERENCES

- Wasserman, L. (2004). *All of Statistics: A Concise Course in Statistical Inference*. New York: Springer, §1.3 (the axioms, p. 5; Theorem 1.8, continuity of probabilities, p. 7), §1.5–§1.7 (independent events, conditional probability, Bayes' theorem).
- Grinstead, C. M., and J. L. Snell (1997). *Introduction to Probability*, 2nd rev. ed. Providence, RI: American Mathematical Society, §4.1 (conditional probability, Bayes' formula and independence; Definitions 4.1–4.5, Theorem 4.1, Exercises 7 and 17).
- Durrett, R. (2019). *Probability: Theory and Examples*, 5th ed. Cambridge University Press, §2.1.4, Theorem 2.1.21 (Kolmogorov's extension theorem, with which §2.1.4 builds an infinite sequence of independent random variables with given distribution functions); numbered as in the author's free PDF.
- Gigerenzer, G., and U. Hoffrage (1995). "How to improve Bayesian reasoning without instruction: Frequency formats." *Psychological Review* 102(4): 684–704, doi:10.1037/0033-295X.102.4.684, Study 1.
- Taiwan Stock Exchange (2023). Index compilation rules, TWSE, IndexS02 (TAIEX and its total-return index, which reinvests cash dividends; revised April 2023), and TWSE, company history (the daily price limit raised from 7% to 10% on 1 June 2015), read 18 October 2026.
- Polars API reference: `Expr.shift` (nulls in the vacated rows), read 18 October 2026.

- Atkinson (7 April 2025). “台股黑色星期一創多項新紀錄，外資抄底轉買超 168.21 億元.” TechNews 科技新報, 財經, 7 April 2025: 936 listed stocks, and 1,702 with the OTC market, closed limit-down that day (the note to Exhibit 6). Read 18 October 2026.
- FinMind open data: `TaiwanStockTotalReturnIndex` (TAIEX), retrieved 18 October 2026.